PEDIATRIC CARDIOVASCULAR DISEASE

Outflow obstruction (well child)

Aortic stenosis

- Narrowing of valve from LV → aorta. **Valvular, subvalvular, or supravalvular** (~5%).
- **Clinical (depends on severity)**: mild–moderate = asymptomatic; **severe = easy fatigability, exertional chest pain, syncope**.
- **Critical infant AS**: survives only if **PDA permits flow** to aorta + coronaries (duct-dependent).

Pulmonary stenosis

- Outflow obstruction RV → pulmonary artery. Mild = asymptomatic; severe → RV overload/failure. [Standard: ejection systolic murmur left upper sternal edge, balloon valvuloplasty if severe.]

Coarctation of aorta

- Narrowing of aorta (juxtaductal). [Standard: **radio-femoral delay, weak femoral pulses, upper limb hypertension, rib notching**; duct-dependent in neonate → **prostaglandin to keep PDA open**; surgery/balloon.]

Patent ductus arteriosus (PDA)

- Persistent ductus → left-to-right shunt. [Standard: **continuous "machinery" murmur**, bounding pulses; treat indomethacin/ibuprofen (close) or surgical/device closure.]

Heart Failure

Definition & physiology

- State where heart can't deliver adequate **cardiac output** for metabolic needs.
- **CO = HR × SV**. SV determinants: **preload (volume), afterload (pressure), contractility**.

Causes

- **Rhythm**: complete heart block, SVT, VT, sinus node dysfunction.
- **Volume overload**: structural (**VSD, PDA, aortic/mitral regurgitation**), anaemia, sepsis.
- **Pressure overload**: aortic/pulmonary stenosis, coarctation, hypertension.
- **Systolic dysfunction**: myocarditis, dilated cardiomyopathy, malnutrition, ischaemia.
- **Diastolic dysfunction**: hypertrophic/restrictive cardiomyopathy, pericarditis, **tamponade**.

Clinical

- **Infants**: feeding difficulty + sweating, poor weight gain, irritability, weak cry, respiratory distress.
- **Children**: fatigue, effort intolerance, anorexia/abdominal pain, dyspnoea, cough, orthopnoea.
- **Signs**: respiratory distress, ↑JVP, **hepatomegaly**, oedema, basal crepitations, cardiomegaly, **gallop rhythm**, holosystolic murmur (mitral/tricuspid insufficiency).

Investigations

- **CXR** (cardiomegaly, pulmonary vascularity), **ECG** (hypertrophy, ischaemia, rhythm), **echo** (ventricular function), blood gas (acidosis), electrolytes (hypoNa, hypoglycaemia).

Treatment

- Treat underlying cause. General: rest, **semi-upright (children) / infant chair**, modify activity.
- **Diet**: ↑calories (up to 24 cal/oz), **no added salt**.

- **Medications**: **diuretics, inotropes, afterload reducers (vasodilators)**.

Kawasaki Disease

- Group of symptoms/signs; **children <5 yrs**. Cause unknown (?infection/toxin/immune). Low family recurrence.
- **Diagnosis**: **fever ≥5 days + 4 of 5**:

1. **Non-purulent red eyes** (bilateral conjunctivitis)
2. **Lip/tongue/mouth changes** (red cracked lips, **strawberry tongue**)
3. **Hand/feet changes** (red/swollen palms+soles → **peeling fingers/toes ~1 wk after fever**)
4. **Body rash** (red, bumpy; groin/diaper area, peels)
5. **Cervical lymphadenopathy** (usually unilateral, >1.5cm)

- **Treatment**: **aspirin + IVIG** (± steroids). No single cure.
- **Complications**: **coronary artery changes + aneurysms** (the danger).
- **Prognosis**: **95% recover completely**; if no coronary damage on echo → complete recovery likely.

Cardiac Arrhythmias

Supraventricular tachycardia (SVT)

- **Most common childhood arrhythmia**. HR **250–300 bpm**. → poor CO, pulmonary oedema. Presents as **HF in neonate/infant**; can cause **hydrops fetalis + intrauterine death**.
- **Re-entry** via accessory pathway. Usually no structural disease (but echo to check).
- **ECG**: narrow-complex tachycardia 250–300; P wave after QRS (retrograde). **WPW**: short PR + **delta wave** + wide QRS (early antegrade conduction).
- **Management (severely ill → restore sinus rhythm)**:

1. Circulatory/respiratory support, correct acidosis.
2. **Vagal manoeuvres** (carotid massage, **ice pack to face**) — ~80% success.
3. **IV adenosine = treatment of choice** (induces AV block, breaks re-entry).
4. **Synchronized DC cardioversion (0.5–2 J/kg)** if adenosine fails.

- **Maintenance**: flecainide or sotalol. **Digoxin** if no delta wave; **propranolol** added if pre-excitation. 90% no further attacks after infancy → **stop treatment at age 1 if no WPW**. WPW relapse/risk → **radiofrequency/cryoablation** of accessory pathway.

Long QT syndrome

- Sudden LOC during **exercise/stress/emotion** (late childhood); may be misdiagnosed as **epilepsy**. Unrecognized → **sudden death from VT**.
- **Autosomal dominant**, many genetic causes; **channelopathy** (Na/K/Ca channels). Associated: **erythromycin**, electrolyte disorders, head injury.
- Assess anyone with FHx sudden unexplained death or syncope on exertion.

Rheumatic Fever

- Multisystem **autoimmune response to group A streptococcus**; children **5–15 yrs**; may progress to RHD.
- Higher in low/middle-income countries (19/100,000) — **household overcrowding**.

Infective Endocarditis

- All CHD (**except secundum ASD**), including neonates, at risk. **Highest risk = turbulent jet** (VSD, coarctation, aorta, PDA) or **prosthetic material**.

- **Presentation**: fever, anaemia/pallor, **clubbing (late)**, **splinter haemorrhages**, **splenomegaly**, **microscopic haematuria**, changing cardiac signs, neurological signs (cerebral infarction), retinal infarcts, arthralgia, necrotic skin lesions.
- **Diagnosis**: **multiple blood cultures BEFORE antibiotics**; **echo (vegetations)** confirms but can't exclude; raised acute phase reactants (monitor).
- **Most common organism: α-haemolytic streptococcus (Strep viridans)**.
- **Treatment**: **high-dose penicillin + aminoglycoside, 6 weeks IV**; surgical removal if infected prosthetic material.

High-yield one-liners

- **Critical AS / coarctation in neonate = duct-dependent → prostaglandin keeps PDA open.**
- **HF: CO = HR × SV (preload/afterload/contractility); infant = feeding difficulty + sweating + poor weight gain.**
- **Kawasaki: fever ≥5d + 4 of 5 (eyes, lips/strawberry tongue, hands/feet peeling, rash, cervical node); IVIG + aspirin; coronary aneurysms.**
- **SVT = commonest childhood arrhythmia, 250–300 bpm; adenosine = treatment of choice; WPW = delta wave.**
- **Long QT = syncope on exertion, mistaken for epilepsy, AD channelopathy, sudden death.**
- **IE: highest risk turbulent jet (NOT secundum ASD); Strep viridans; blood cultures before antibiotics; penicillin + aminoglycoside 6 wks.**

PEDIATRIC CARDIOVASCULAR DISEASE — Revision Table

Bold = high-yield. [bracket] = standard added (slide image-only).

OUTFLOW OBSTRUCTION (well child)

Lesion	Key points
Aortic stenosis	Valvular/subvalvular/supravalvular (5%). Severe → **fatigue, exertional chest pain, syncope**. **Critical infant AS = duct-dependent (PDA keeps coronary/aortic flow)**
Pulmonary stenosis	RV outflow obstruction; severe → RV failure. [ESM left upper sternal edge; balloon valvuloplasty]
Coarctation of aorta	[**Radio-femoral delay, weak femoral pulse, UL hypertension, rib notching**; neonate duct-dependent → **prostaglandin**]
PDA	[**Continuous machinery murmur**, bounding pulses; indomethacin/ibuprofen or device closure]

HEART FAILURE

Aspect	Key points
Definition	Heart can't meet metabolic CO needs. **CO = HR × SV** (preload, afterload, contractility)
Causes	Rhythm (HB, SVT, VT); **volume overload (VSD, PDA, regurgitation)**, anaemia, sepsis; pressure overload (AS, PS, coarctation, HTN); systolic (myocarditis, DCM); diastolic (HCM, RCM, tamponade)
Clinical infant	**Feeding difficulty + sweating, poor weight gain, irritability, respiratory distress**
Clinical child	Fatigue, effort intolerance, anorexia/abdo pain, dyspnoea, orthopnoea
Signs	↑JVP, **hepatomegaly**, oedema, basal creps, cardiomegaly, **gallop**, holosystolic murmur
Ix	CXR (cardiomegaly), ECG, **echo (function)**, blood gas (acidosis)
Treatment	Treat cause; semi-upright/infant chair; ↑**calorie + no added salt**; **diuretics, inotropes, afterload reducers**

KAWASAKI DISEASE (children <5yr)

Aspect	Key points
Diagnosis	**Fever ≥5 days + 4 of 5**: 1. Non-purulent red eyes 2. **Lips/strawberry tongue** 3. Hands/feet red→**peeling** 4. Body rash (groin) 5. **Cervical node (unilateral >1.5cm)**
Treatment	**Aspirin + IVIG** ± steroids
Complication	**Coronary artery changes + aneurysms**
Prognosis	**95% complete recovery** if no coronary damage on echo

ARRHYTHMIAS

Arrhythmia	Key points
SVT	**Commonest childhood arrhythmia**; HR **250–300**; re-entry; HF in infant, hydrops fetalis. **WPW = short PR + delta wave**. Mgmt: vagal (ice/face 80%) → **IV adenosine (TOC)** → DC cardioversion (0.5–2 J/kg). Maintenance flecainide/sotalol; **ablation if WPW relapse**
Long QT	Syncope on **exertion/stress** (mistaken for epilepsy); **AD channelopathy** (Na/K/Ca); sudden death from VT; assoc erythromycin/electrolytes

RHEUMATIC FEVER & INFECTIVE ENDOCARDITIS

Topic	Key points
Rheumatic fever	Autoimmune to **group A strep**; age 5–15; → RHD; overcrowding/low-income
Infective endocarditis	All CHD risk **except secundum ASD**; **highest = turbulent jet (VSD, coarctation, PDA) / prosthetic**. Signs: fever, splinter haemorrhages, **splenomegaly, microscopic haematuria**, clubbing (late). **Blood cultures BEFORE antibiotics**; echo vegetations; **Strep viridans** commonest; **penicillin + aminoglycoside 6 wks IV**